Morgan Wang's Portfolio

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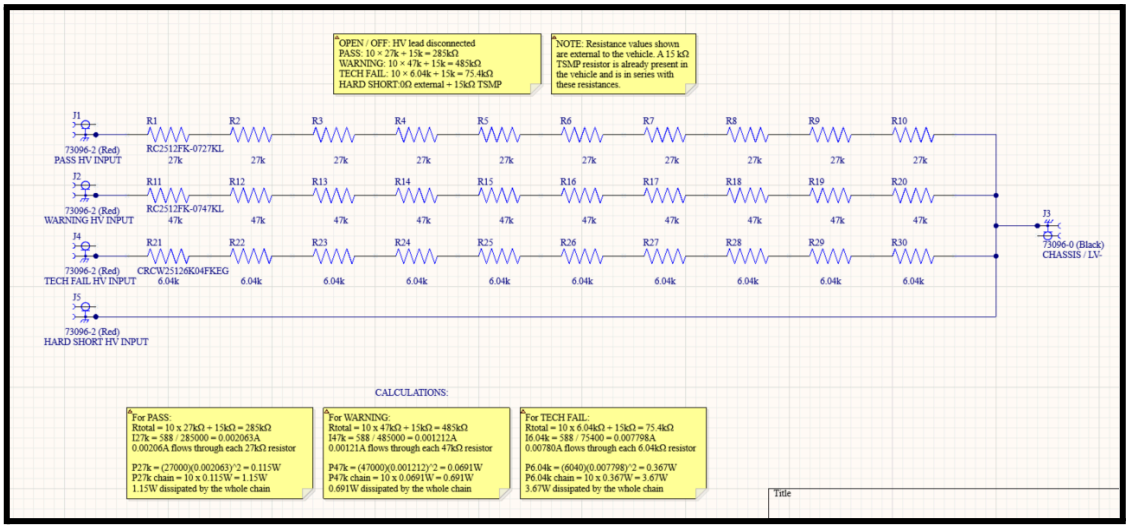
IMD Test Box - UW Formula Electric

What: Developing a passive insulation monitoring fault injection test box for the Formula Electric vehicle, interfacing through the HV+/HV- Tractive System Measuring Points (TSMP) and chassis reference for controlled test conditions.

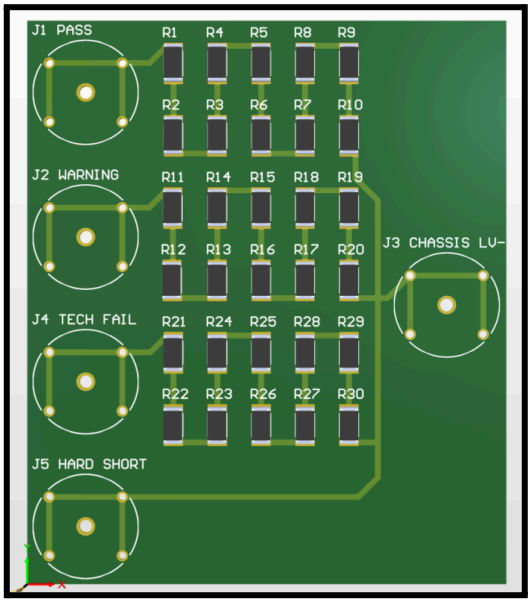
How: Created Altium Designer schematics and resistor networks. Accounted for 15 kOhm TSMP protection resistance and evaluated a 588 V tractive system against the FSAE 250 Ohm/V insulation criterion.

Result: Calculated a 147 kOhm critical insulation threshold and design target equivalent fault resistances of 285, 485, and 75 kOhm. Planned selection states include PASS, WARNING, TECH FAIL, and a hard short scenario. The design provides a repeatable method to investigate firmware reported warnings and vehicle shutdown responses.

Next steps: Physical testing on the car after the PCB has been printed.



Altium Schematic



IMD Test Box PCB Render

Position	Function	Target total fault (R_fault)	Box resistor (R_box)	Worst case box power (P_box)	Expected IMD response
1	OPEN/OFF	∞ (disconnected)	Open circuit	0W	Normal operation (OK_hs high). Safe position for connecting leads
2	PASS	≈285kΩ	270kΩ	0.85W	Car stays on: Well above 147 kΩ; no false tripping
3	WARNING	≈500kΩ	470kΩ - 485kΩ	0.48W	CAN Warning: Flags warning status over CAN; car stays on
4	TECH FAIL	75kΩ (50% rule margin)	60kΩ	3.7W	Mandatory Trip: OK_hs drops to ≤0.2V; SDC must open in ≤30s
5	HARD SHORT	15kΩ (sees only R_bpr)	0Ω (direct wire)	0W (car BPR takes 23W)	Immediate Trip: Direct HV-to-chassis short simulation

Test Conditions and Vehicle Response

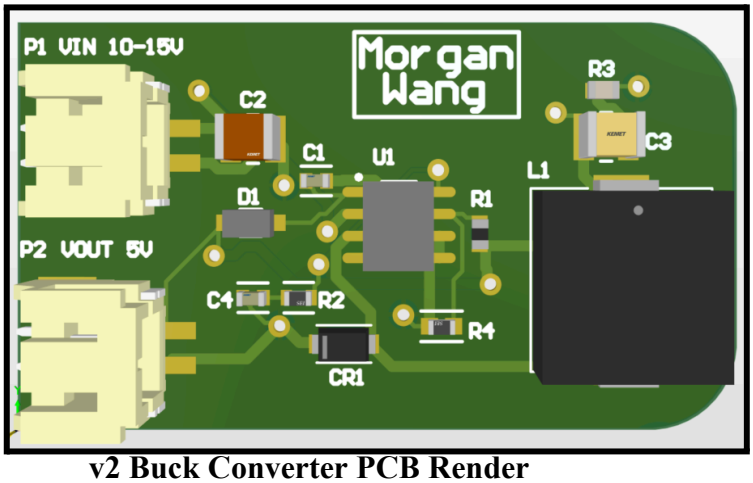
5V DC-DC Buck Converter - UW Midnight Sun

What: Designed a compact 10-15 V input to regulated 5 V, 1 A switching buck converter using the LT1375 for a low voltage power supply design task.

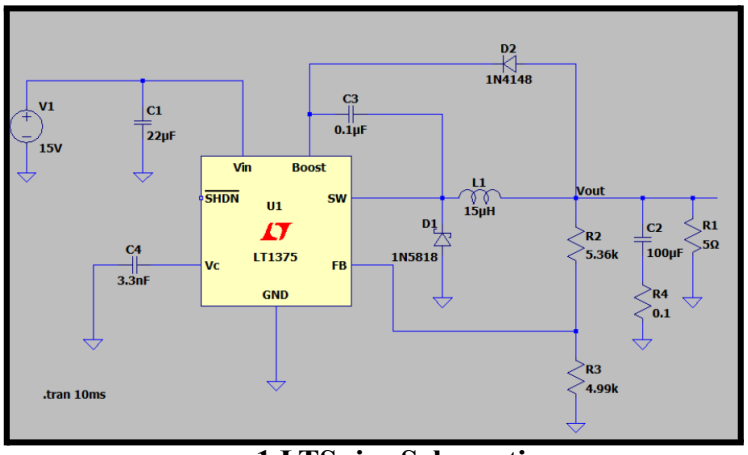
How: Selected a 15 uH inductor, Schottky freewheeling diode, input/output capacitors, and resistive feedback components using datasheets and calculations. Simulated switching behavior and component trade offs in LTspice.

Result: Reduced simulated output ripple from 45.7 mVpp (v1) to 23.6 mVpp (48.3%, v2) by changing output capacitor ESR from 0.10 to 0.05 Ohm and tuning compensation (1 kOhm / 3.3 nF) at 15 V input and 1 A load.

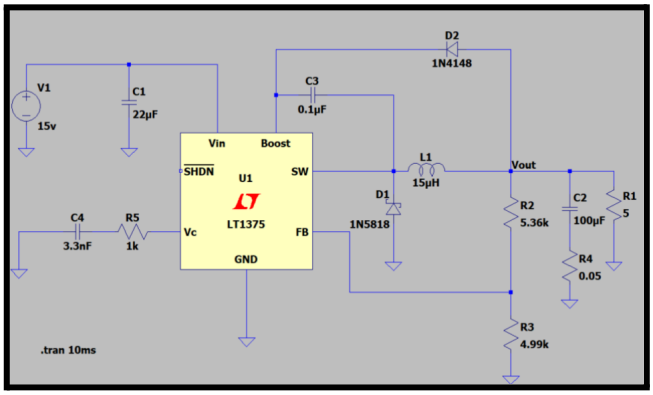
PCB: Designed a 41.4 x 23.6 mm two layer SMD layout in Altium Designer with 25 mil power traces, 10 mil signal traces, a bottom layer GND polygon, and via connected returns. Reported zero violations under the DRC rules.



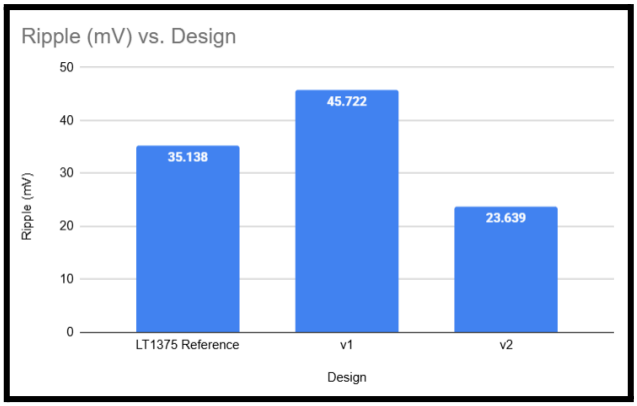
v2 Buck Converter PCB Render



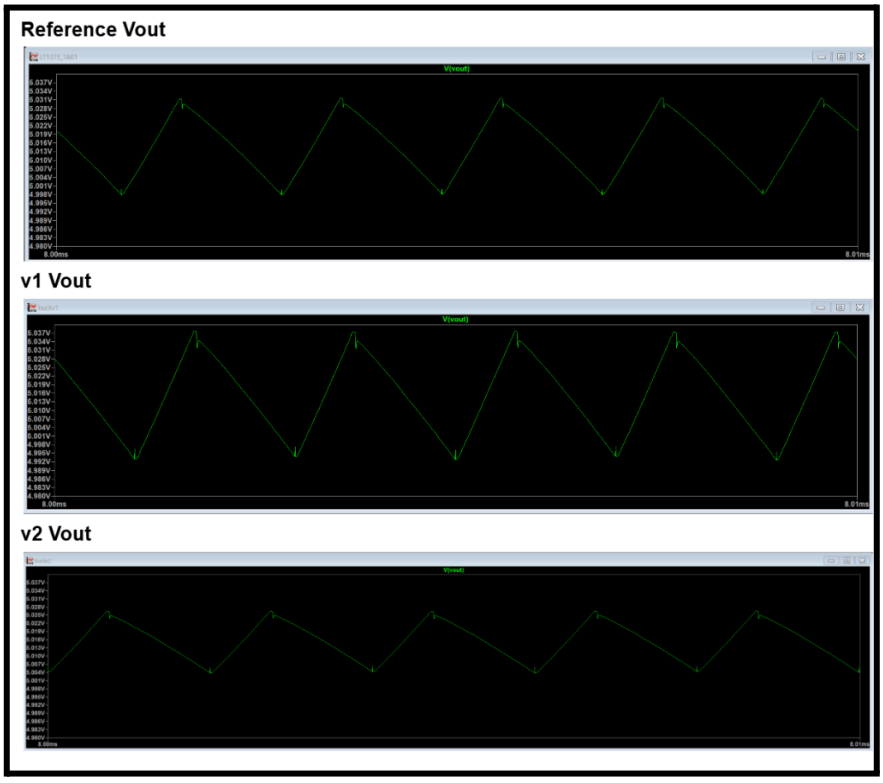
v1 LTSpice Schematic



v2 LTSpice Schematic



Simulated output ripple comparison



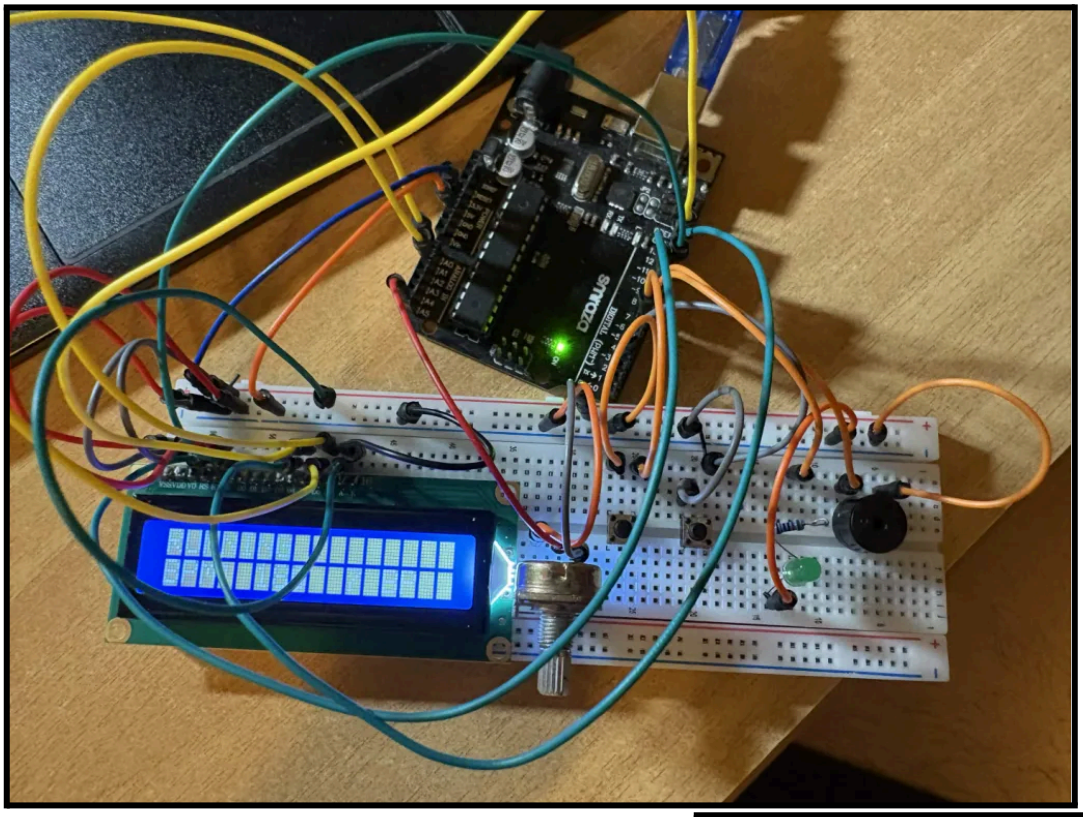
Output Voltage Comparison (same time interval & y-axis, 5.04V-4.98V)

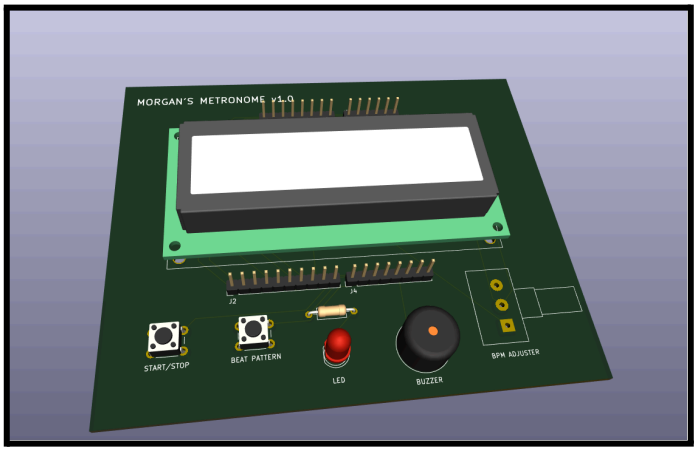
Morgan's Metronome v1.0 - Personal Project

What: Built an Arduino UNO based digital metronome after finding myself getting distracted using my phone while practicing saxophone. Designed it to help maintain a consistent rhythm, featuring adjustable 40–180 BPM tempo, selectable beat patterns, pushbutton controls, a potentiometer, and a 16×2 LCD.

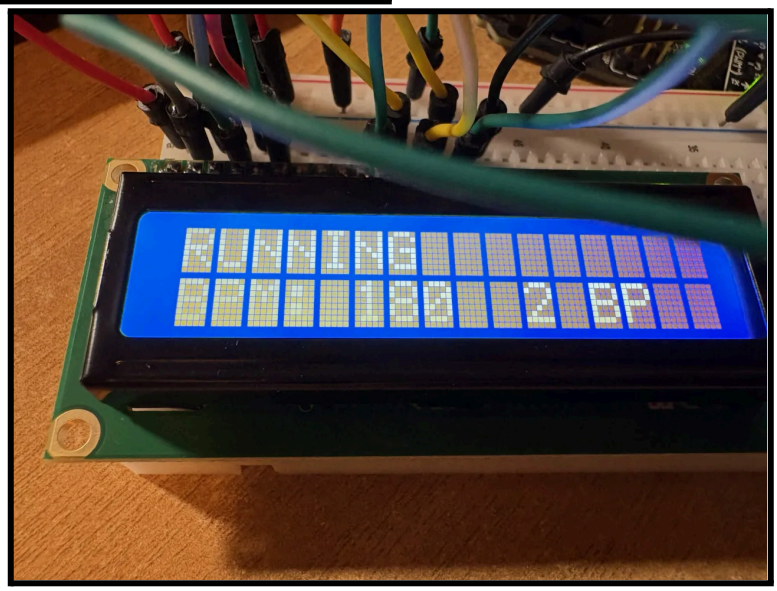
How: Programmed the timing and user interface in C++. Used millis() based non blocking timing to schedule beats while keeping tempo and button controls responsive. Prototyped and debugged the circuit on a breadboard before designing a custom Arduino UNO shield PCB in KiCad to integrate the components.

Result: Developed a functional metronome with an audible buzzer and visual LED beat indicators, adjustable tempo, and selectable rhythm patterns. PCB is on its way!





KiCad Arduino Shield 3D Render



1602A LCD display